

Database Description

Entities & Attributes

1. Employee

- Employee id (pk)
- Password
- Position
- Salary
- Status
- Last_Login

2. Incharge

- Employee id (pk)
- Name
- Operating stage
- Email
- Address
- Contact Number (Multivalued)

3. Worker

- Employee id (pk)
- Name
- Address
- Grade
- Email
- Contact Number (Multivalued)

4. Production Stage

- Stage id (pk)
- Stage Name
- Stage Progress
- Assigned Workers
- Start Date
- End Date

5. Machinery

- Machine id (pk)
- Name
- Type
- Cost per unit
- Quantity

6. Costing

- Costing id (pk)
- Final Bill

7. Order

- Order id (pk)
- Description
- Order date
- Estimate Delivery

8. Material

- Material id (pk)
- Name
- Type
- Unit of measure
- Unit price

9. Inspection

- Inspection id (pk)
- Passed Quantity
- Failed Quantity
- Remarks

10. Final Product

- Final Product id (pk)
- Grade
- Lot Number
- Date of Completion

11. Packaging

- Package id (pk)
- Date
- Weight per pack
- Quantity per Pack
- Total package
- Type

12. Shipment

- Shipment id (pk)
- Tracking number
- Estimated delivery
- Destination
- Shipped Date

13. Buyer

- Buyer id (pk)
- Name
- Brand
- Address
- Account No
- Email
- Contact Number (Multivalued)

14. Payment

- Payment id (pk)
- Total Amount
- Paid amount
- Remaining amount
- Payment Method
- Date

15. Accounts

- Transaction id
- Status (Credited / Debited)
- Amount
- Date
- Associated bank

16. BOM

- BOM id (pk)
- Material description
- Unit Bill
- Total Bill

17. Supplier

- Supplier id (pk)
- Name
- Address
- Email
- Contact Number (Multivalued)

18. Order_Style (Weak Entity of Order)

- Style id (pk)
- Style name
- Color
- Size

Relationships & Attributes

1. Employee IS-A Incharge , Worker (Disjoint)
2. Rel_Worker_ProductionStage
 - Worker . Employee id (fk)
 - Stage id (fk)
3. Rel_Incharge_Rel_Worker_ProductionStage (Aggregation)
 - Incharge . Employee id (fk)
 - Worker . Employee id (fk)
 - Stage . Id (fk)
 - Operation Progress
4. Rel_ProductionStage_Machinery
 - Stage id (fk)
 - Machine id (fk)
 - Used Duration
 - Used Cost
5. Rel_Machinery_Costing
 - Machine id (fk)
 - Costing id (fk)
 - Machinery total cost (derived)
6. Rel_Costing_Order
 - Costing id (fk)
 - Order id (fk)
7. Rel_Order_Material
 - Order id (fk)
 - Material id (fk)
8. Rel_ProductionStage_Inspection
 - Stage id (fk)
 - Inspection id (fk)
9. Rel_Inspection_FinalProduct
 - Inspection id (fk)
 - Final Product id (fk)
10. Rel_FinalProduct_Packaging
 - Final Product id (fk)
 - Package id (fk)
11. Rel_Packaging_Shipment
 - Package id (fk)
 - Shipment id (fk)

12. Rel_Shipment_Buyer
 - Shipment id (fk)
 - Buyer id (fk)
13. Rel_Buyer_Payment
 - Buyer id (fk)
 - Payment id (fk)
14. Rel_Buyer_Order
 - Order id (fk)
 - Buyer id (fk)
15. Rel_Costing_Payment
 - Costing id (fk)
 - Payment id (fk)
 - Profit Margin
16. Rel_Payment_Accounts
 - Payment id (fk)
 - Transaction id (fk)
17. Rel_Accounts_Employee
 - Transaction id (fk)
 - Employee id (fk)
 - Position
 - Max Salary
 - Min Salary
18. Rel_Costing_BOM
 - Costing id (fk)
 - BOM id (fk)
19. Rel_Accounts_BOM
 - Transaction id (fk)
 - BOM id (fk)
20. Rel_Accounts_Supplier
 - Transaction id (fk)
 - Supplier id (fk)
21. Rel_Supplier_BOM_Material (Ternary)
 - Supplier id (fk)
 - BOM id (fk)
 - Material id (fk)
 - Time required
 - Quantity

22. Rel_Order_OrderStyle (Total participation)

- Order id (fk)
- Style id (fk)
- Quantity

[Pk - primary key

Fk - foreign key]